Hongren Zhu

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Education

The University of Hong Kong Hong Kong
Master of Statistics Expected 2027

- **Current coursework:** Financial Data Analysis; Spatial Data Analysis; Fundamentals of Statistical Inference; Time Series Forecasting; Marketing Analytics.

Dalhousie University Halifax, NS, Canada
Bachelor of Science in Statistics, First-Class Honours Jun. 2025

- **GPA:** 3.98/4.3. **Selected coursework:** Statistical Learning with R (A+), Optimization (A+), Design of Experiments (A+).

Work Experience

Shandong Yimeng Traffic Development Investment Group Co., Ltd.
Data Analyst Assistant Jul. 2025 – Aug. 2026

- Collected accident, near-miss and safety-violation records for a 114.3 km expressway section and managed incident data in a SQL database.
- Analysed accident frequency using ARIMA and Getis–Ord statistics; mapped hotspots in Tableau to identify high-risk locations and periods.
- Produced analysis reports recommending traffic-sign and surveillance-camera placement to support road-safety planning.

Linyi Electric Power Industry Co., Ltd., State Grid Corporation of China
Intern, Safety Supervision Department Jul. 2024 – Sep. 2024

- Supported safety supervision for power-line, transformer and distribution construction projects at 220 kV and below.
- Inspected construction sites, identified safety-protocol violations and worked with site managers on corrective action.
- Developed a systematic approach to project planning and control to support compliance with safety regulations and operational standards.

Research & Projects

Neural Network Loss Functions and Optimization (MNIST) Jan. 2025 – Jun. 2025
Research Paper

- Built PyTorch convolutional neural networks and evaluated nine loss-function and optimizer combinations under controlled hyperparameters, comparing accuracy, F1-score, convergence and training stability.
- Identified focal loss with stochastic gradient descent as the best-performing combination: **99.16% test accuracy** and **0.9916 F1-score**; visualised learning curves and confusion matrices.

Neural Network Architecture Comparison (CIFAR-10) Nov. 2024 – May 2025
Honours Thesis, Dalhousie University

- Implemented five neural network architectures in PyTorch; preprocessed and augmented image data, tuned parameters and trained models on GPUs.
- Compared accuracy, convergence and resource efficiency: ResNet achieved **93.68% accuracy**, while the convolutional neural network offered the best parameter efficiency.

Biodiversity Analysis, Musquash Marine Protected Area Sep. 2024 – Jan. 2025
Research Project, Dalhousie University

- Derived biodiversity indices from 2010–2023 Department of Fisheries and Oceans data and examined relationships with sediment composition and environmental factors.
- Modelled seasonal and spatial variation using linear regression, generalised additive models and regression trees; identified sediment texture and organic content as key explanatory variables.

Skills

Programming: Python, R, SQL (MySQL)

Machine learning: PyTorch

Tools: Excel, Tableau, Git, JupyterLab, RStudio, LaTeX

Languages: Mandarin (native), English (professional proficiency)